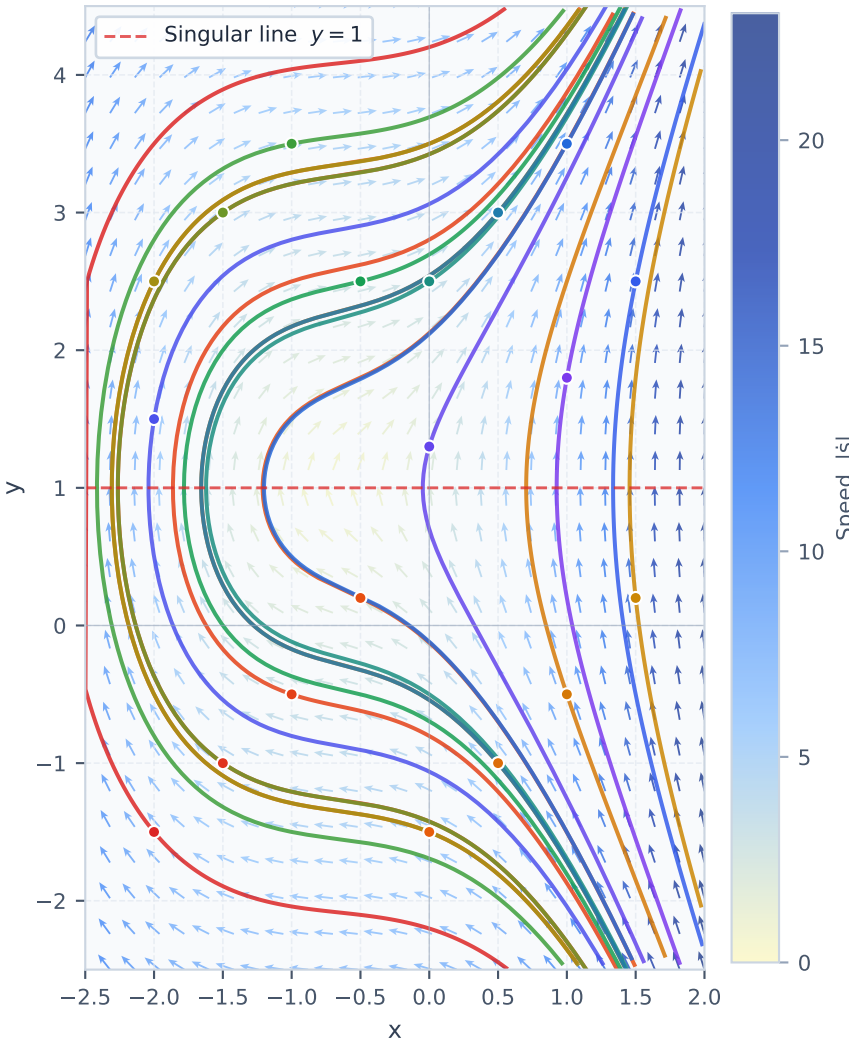
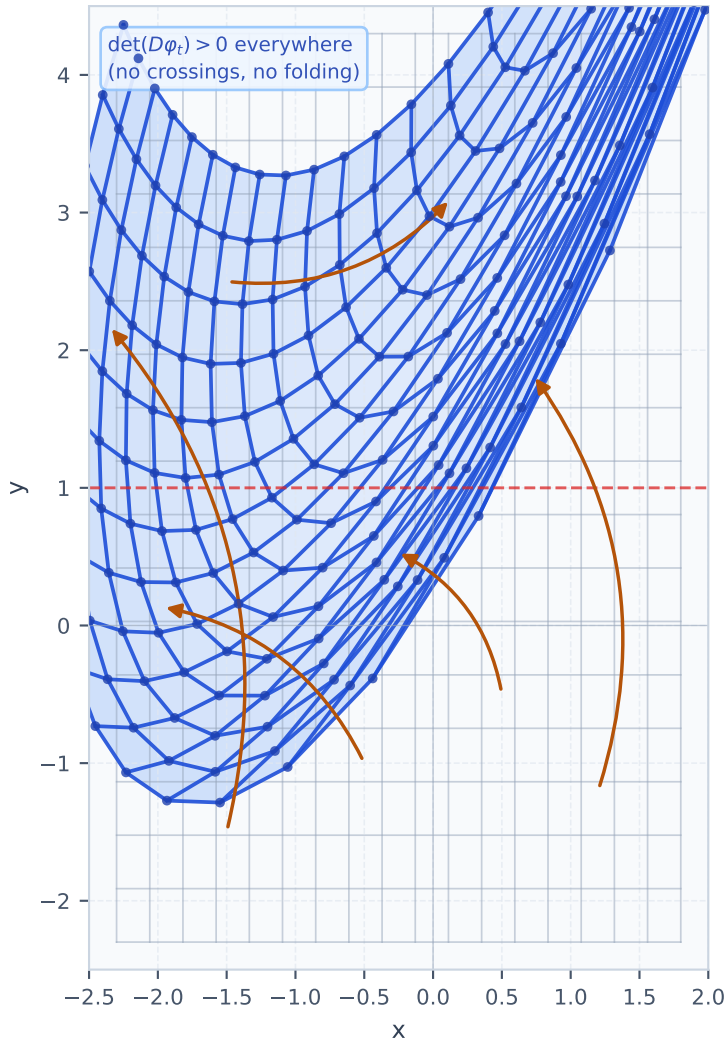


Flow Map: $\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y - 1)} \iff \dot{x} = 2(y - 1), \quad \dot{y} = 3x^2 + 4x + 2$

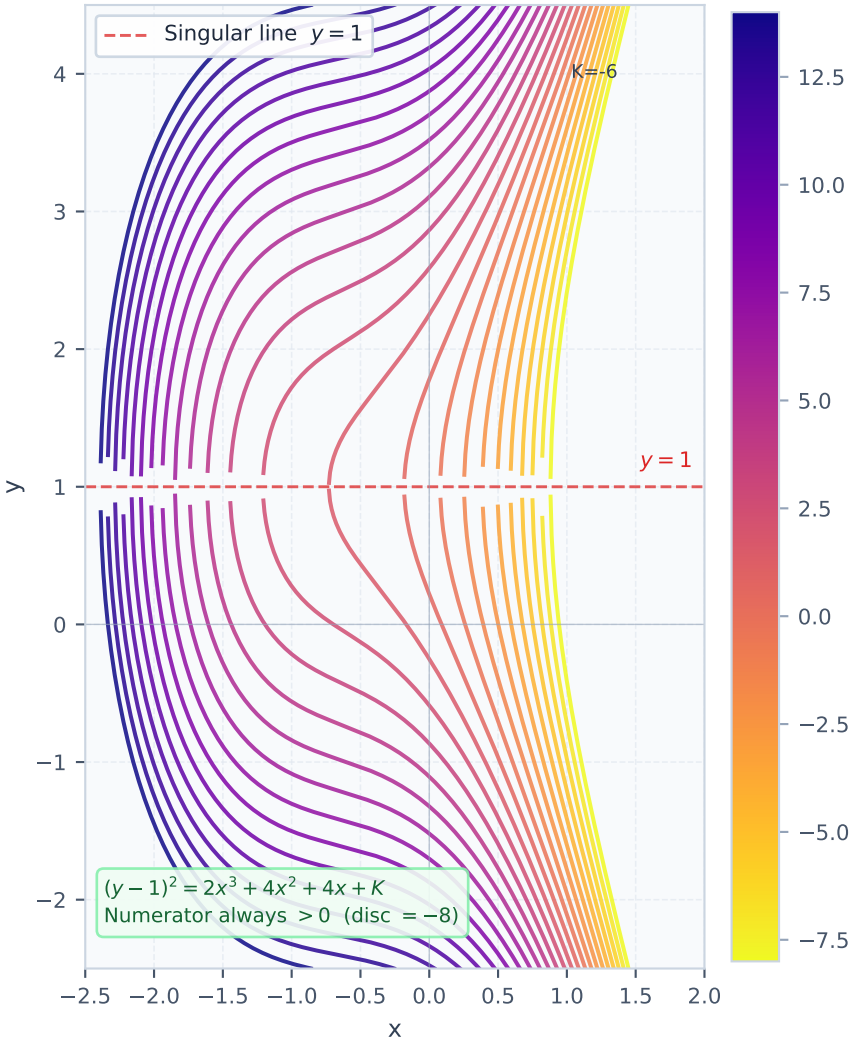
Phase Portrait — Vector Field & Integral Curves



Deformed Grid — Diffeomorphism φ_t ($t = 0.45$)



Solution Family: $(y - 1)^2 = 2x^3 + 4x^2 + 4x + K$



Jacobian $\det(D\varphi_t)$ — Confirms Diffeomorphism

